[S07] Retrieval-Augmented Generation (RAG)

Grounding LLM Responses in Documents: An Empirical Study on HCMUS Regulatory Data

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CS420 — Introduction to Artificial Intelligence

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Outline

- **Theoretical Foundation & Modeling** (Speaker 1: Nam Kiệt, 00:00 – 05:00)
  Parametric vs. non-parametric memory, RAG pipeline, sequence vs. token formulations
- **HCMUS Corpus & Retrieval Evaluation** (Speaker 2: Lê Huy, 05:00 – 10:00)
  Regulatory corpus (51 pages), TF-IDF vs. BM25 vs. BGE-M3, worked example Q01
- **Answer Quality, Failure Analysis & Verification** (Speaker 3: Tuấn Lĩnh, 10:00 – 15:00)
  Controlled 4-way evaluation, 5 failure archetypes, hallucination analysis, verification loop

CS420 Seminar S07: 15-minute presentation balanced across 3 team members (5:00 each) + 5-minute Q&A.

Does providing external documents eliminate hallucinations?

- **Parametric Memory Limits:** LLM weights encode knowledge statically, leading to factual hallucinations, cut-off staleness, and unverified assertions.
- **Non-Parametric Grounding:** An external document index provides modular, updatable, and inspectable evidence for factual verification.
- **The Core Research Question:** Does conditioning generation on retrieved passages eliminate hallucinations, or does it shift error points to the retrieval pipeline?
- **Seminar Objective:** Combine probabilistic modeling (Lewis et al., 2020) with an empirical audit on 4 official HCMUS academic regulations.

Lewis et al. (NeurIPS 2020), §1; Schneider (2024), GenXAI, §1.

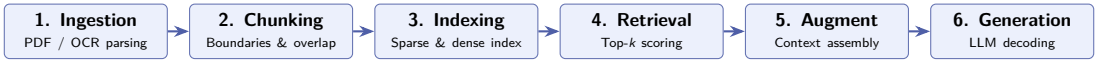
Knowledge Grounding Paradigms

Paradigm	Operating Mechanism	Latency / Compute	Knowledge Update
Fine-Tuning	Updates parametric neural weights	High training cost	Requires retraining
Long-Context	Appends all source text into prompt	Quadratic token cost	Dynamic per query
RAG (Hybrid)	Pairs top-k retrieval with LLM generation	Bounded context cost	Hot-swappable index

- **Decoupled Architecture:** Parametric model acts as a language synthesizer; non-parametric index acts as the verifiable knowledge store.
- **Efficiency & Precision:** Mitigates "Lost-in-the-Middle" phenomena and context dilution by filtering large corpora into top-k evidence.
- **Lewis et al. Finding:** RAG-Sequence achieved 44.5 Exact Match on Natural Questions, outperforming a 50$\times$ larger parametric T5-11B model (36.6).

Lewis et al. (NeurIPS 2020), Table 1; Liu et al. (2024), TACL.

RAG Architecture & Theoretical Failure Points



- **Pre-Retrieval Failure Points:** OCR noise on scanned documents; premature text splits detaching qualifying exceptions from primary articles.
- **Retrieval Failure Points:** False negative omissions (gold threshold clauses rank outside top-k); lexical distractor confusion.
- **Post-Retrieval Failure Points:** Context distraction; condition omission; table column misreading; failure to produce verifiable citation IDs.

Adapted from Lewis et al. (2020), Figure 1; CS420 S07 Report §4.2, Figure 1.

Probabilistic Modeling: Sequence vs. Token

1. RAG-Sequence (Sequence-Level Marginalization):

$$p_{\text{Seq}}(y \mid x) \approx \sum_{z \in \text{top}-k} p_\eta(z \mid x) \prod_{i=1}^{T} p_\theta(y_i \mid x, z, y_{<i})$$

2. RAG-Token (Token-Level Step Marginalization):

$$p_{\text{Tok}}(y \mid x) \approx \prod_{i=1}^{T} \left[\sum_{z \in \text{top}-k} p_\eta(z \mid x) p_\theta(y_i \mid x, z, y_{<i}) \right]$$

- **Latent Passage (z):** Retrieved documents act as latent variables marginalized across top-k candidates.
- **Sequence Formulation:** Generates entire response y from a single document z; preserves topic coherence.
- **Token Formulation:** Dynamically switches documents across tokens; ideal for synthesizing multi-source facts.
- **Variables:** x: query; $y_{<i}$: generated prefix; η, θ: retriever/generator parameters; $p \in [0, 1]$.

Lewis et al. (NeurIPS 2020), §2.1, Equations (1)–(2).

Worked Numerical Example ($k = 2, T = 2$)

Passage (z)	Retriever Weight $p_\eta(z \mid x)$	$p_\theta(a \mid x, z)$	$p_\theta(b \mid x, z, a)$
z_1	0.70	0.80	0.90
z_2	0.30	0.40	0.20

- **RAG-Sequence Calculation:**

$$p_{\text{Seq}} = 0.70 \times (0.80 \times 0.90) + 0.30 \times (0.40 \times 0.20) = 0.5040 + 0.0240 = \mathbf{0.5280}$$

- **RAG-Token Calculation:**

$$p_{\text{Tok}} = (0.70 \times 0.80 + 0.30 \times 0.40) \times (0.70 \times 0.90 + 0.30 \times 0.20) = 0.6800 \times 0.6900 = \mathbf{0.4692}$$

- **Operator Inversion:** The discrepancy (0.5280 vs. 0.4692) reflects the non-commutativity of $\sum$ and $\prod$.
- **Implementation Distinction:** Modern instruction-tuned QA (e.g., Qwen) performs prompt conditioning rather than explicit marginalization.

CS420 S07 Report §3.2, Table 1. Probabilities parameterized to demonstrate mathematical formulation.

Experimental Corpus: HCMUS Regulatory Documents

ID	Regulatory Document Title and Reference	Scope	Pages
D01	Undergraduate Academic Regulations 2026 (Dec. 3087/QĐ-KHTN)	Academic Rules	21
D02	Student Conduct Assessment 2020 & Amendment 2023	Conduct Scoring	16
D03	Tuition Fee Rates AY 2026–2027 (Dec. 3317/QĐ-KHTN)	Fees (K2022+)	5
D04	Foreign Language Graduation Standards (Dec. 2464/QĐ-KHTN)	Exit Standards	9
Total	Authoritative four-document regulatory snapshot	PDF Corpus	**51**

- **Corpus Scale:** 51 pages segmented into 80 parent sections and 162 chunks under an 800-token ceiling (`cl100k_base`).
- **Evaluation Benchmark:** 20 administrative queries mapped to 30 gold evidence groups (held out from indexing).
- **Metadata Preservation:** Articles, PDF page numbers, cohort scopes, and table footnotes injected into chunk headers.
- **Scope Distinction:** Exceeds standard 5–10 page guideline to test real-world administrative cross-referencing.

CS420 S07 Report §4.1, Table 2. Verified against official HCMUS portals (commit 3bb5f3a).

Retrieval Configurations & Evaluation Protocol

Retriever	Type	Representation	Hyperparameters
TF-IDF	Sparse Lexical	Unigram + bigram frequencies	Sublinear TF, cosine distance
BM25	Sparse Lexical	Term frequency saturation & length penalty	Vietnamese tokens, $k_1 = 1.5, b = 0.75$
BGE-M3	Dense Neural	1024-dim multilingual dense embeddings	Normalized vectors, cosine similarity

- **Evaluation Metrics (Macro-averaged across 20 queries, 30 gold evidence groups):**
  - **Evidence Recall@k:** Fraction of necessary gold evidence groups retrieved (r/m).
  - **Hit Rate@k:** Binary indicator if at least one gold evidence group is found ($r \geq 1$).
  - **All Evidence Hit@k:** Binary indicator if all gold evidence groups are retrieved ($r = m$).
  - **MRR@k:** Reciprocal rank of the first relevant evidence chunk (0 if not in top-k).
- **Benchmark Cut-offs:** Evaluated at $k \in \{1, 3, 5\}$ to compare early precision vs. multi-hop recall.

CS420 S07 Report §4.1; Retrieval evaluation scripts at commit 5771d80.

Experiment 1: Retrieval Performance

Retriever	Recall@1	Recall@3	Recall@5	Hit@5	All Evidence@5	MRR@5
TF-IDF	70.83%	82.50%	82.50%	90.00%	15/20	0.8250
BM25	**80.00%**	**84.17%**	84.17%	90.00%	**16/20**	**0.8667**
BGE-M3	65.00%	81.67%	**89.17%**	**95.00%**	**16/20**	0.7867

- **Early Precision (BM25):** BM25 dominates top-1 ranking (Recall@1 = 80.00%) via exact lexical matching of regulatory terms.
- **High-Cutoff Coverage (BGE-M3):** Dense embeddings lead at $k = 5$ (Recall@5 = 89.17%, Hit@5 = 95.00%) by capturing semantic paraphrases.
- **Plateau Effect:** Lexical methods saturate at $k = 3$, whereas BGE-M3 gains +7.50 percentage points expanding from $k = 3$ to $k = 5$.
- **Crucial Takeaway:** **Hit@5 = 95.00% does NOT imply 95% answer accuracy!** Downstream generation bottlenecks persist.

CS420 S07 Report §5, Table 4. Evaluated on 20 queries, 30 gold evidence groups (commit 5771d80).

Worked Example (Q01): Top-5 Candidate Chunks

Benchmark Query (Q01): *According to the 2026 Academic Regulations, how does a prerequisite course differ from a pre-learning course?*

Rank	Chunk Identifier	Cosine	Page	Content Summary and Regulatory Scope
1	`D01_regulation_article_3_03`	**0.6271**	4	**Gold Evidence:** Article 3, Clause 3(c) (Both definitions)
2	`D01_regulation_article_27_01`	0.5813	21	Article 27: Transitional clauses for earlier cohorts
3	`D01_regulation_article_3_02`	0.5704	3	Article 3, Clause 2: Elective and general course definitions
4	`D01_regulation_article_6_02`	0.5643	7	Article 6: Semester scheduling and course limits
5	`D01_regulation_article_18_02`	0.5540	16	Article 18: Credit exemptions and transfer guidelines

- **Successful Top-1 Alignment:** Rank-1 passage provides 100% of required gold evidence for both course concepts.
- **Crucial Regulatory Distinction:** Prerequisite requires "passing" the exam; pre-learning only requires "completing the coursework".
- **Distractor Risk:** Ranks 2–5 introduce adjacent credit/scheduling rules that risk inducing model confusion.

CS420 S07 Report §4.2, Table 3. Dense ranking log from `retrieval_hits.json`.

Worked Example (Q01): Grounded vs. Baseline Output

Configuration	Score	Generated Response Content & Attribution
Baseline (No RAG)	0 / 2	Hallucinates generic academic progression; falsely claims both course types require passing grades; lacks any regulatory citation.
Grounded RAG (BGE-M3)	**2 / 2**	Correctly states prerequisite requires **"passing"** (thi đạt), while pre-learning only requires **"completing the study process"**. [D01_regulation_article_3_03]*

- **Prompt Augmentation:** Context injected under structured SOURCES block with chunk IDs, article titles, and page numbers.
- **Factual Grounding Contrast:** Parametric memory guesses incorrectly; RAG context enables precise legal differentiation.
- ***Human Attribution Caveat:** Qwen generated the correct text but failed to emit the citation ID; the reference tag was verified and added post-hoc by human auditors.

CS420 S07 Report §4.2; Task 6 answer comparison logs. *Citation verified and appended during human audit.

Experiment 2: 4-Way Answer Quality

Configuration	Fully Correct	Partial	Incorrect	Unsupported	Valid Citations
Baseline (No RAG)	0/20 (0.0%)	3/20 (15.0%)	17/20 (85.0%)	15/20 (75.0%)	0/20 (0.0%)
RAG (TF-IDF)	**10/20 (50.0%)**	5/20 (25.0%)	5/20 (25.0%)	8/20 (40.0%)	0/20 (0.0%)
RAG (BM25)	9/20 (45.0%)	6/20 (30.0%)	5/20 (25.0%)	7/20 (35.0%)	0/20 (0.0%)
RAG (BGE-M3)	**10/20 (50.0%)**	6/20 (30.0%)	4/20 (20.0%)	**6/20 (30.0%)**	0/20 (0.0%)

+45.0 to +50.0 pp accuracy gain over unaugmented baseline (0% → 50%)

- **Recall ≠ Accuracy Bottleneck:** Despite higher Evidence Recall@5 (89.17%), BGE-M3 ties with TF-IDF at 50% accuracy.
- **Significant Hallucination Drop:** Unsupported claims decrease from 75% in baseline to 30% in BGE-M3 and 35% in BM25.
- **Universal Citation Blindness:** All 60 RAG answers omitted formal citation tags despite explicit prompt constraints.

CS420 S07 Report §5, Table 5. Evaluated on Qwen2.5-1.5B-Instruct across 80 completions (commit `3bb5f3a`).

Empirical Failure Modes: 5 Representative Cases

Query	Observed Prompt and Generation Behavior	Root Cause and Remediation Stage
Q07	Top-5 misses the 65–79 score band chunk. BGE-M3 fabricates "Average 50 points" for a 68-point score.	**Retrieval Omission:** Gold threshold missing; model guesses parametrically.
Q01	Correct definition ranks 1. TF-IDF/BM25 rank English prerequisite at rank 2, derailing output focus.	**Distractor Interference:** Lexical match confuses cross-attention context.
Q05	Model receives Article 19 but asserts credit completion alone suffices for graduation.	**Condition Pruning:** Generator overlooks secondary administrative rules.
Q13	All retrievers capture tuition table. Model reads column C1 (488k) instead of C5 (1,014k vs. 974k).	**Tabular Misalignment:** 1.5B LLM struggles with row-column intersections.
Q19	Model captures valid deadlines but fabricates a non-existent decision suffix identifier.	**Metadata Hallucination:** Generates plausible-sounding official numbers.

- **Comprehensive Distribution:** Failures occur across retrieval (Q07, Q01), reasoning (Q05), tables (Q13), and citations (Q19).
- **High Risk of Structured Data:** Multi-column tables and multi-condition checklists require specialized extraction pipelines.

CS420 S07 Report §6, Table 6. Case analysis from 80-answer generation logs (commit `3bb5f3a`).

Discussion: Does RAG Eliminate Hallucinations?

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- **Paradigm Shift in Errors:** RAG resolves the parametric knowledge deficit (jumping from 0% to 50% accuracy), but introduces pipeline-dependent vulnerabilities.
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- **The Grounding Fallacy:** Supplying relevant documents is a *necessary* condition, but *not a sufficient* condition to guarantee factual truthfulness.
- **Cascading Pipeline Errors:** Hallucinations now emerge from retrieval omissions, adjacent distractors, condition pruning, and tabular distortion.
- **Literature Corroboration:** Aligns with Schneider (2024, GenXAI) and Niu et al. (ACL 2024, RAGTruth): LLMs frequently assert ungrounded claims despite prompt grounding.

Schneider (2024), GenXAI, §1–§2; Niu et al. (ACL 2024), RAGTruth; CS420 S07 Report §1, §6.

Limitations & Applicability Boundaries

- **Benchmark Sample Size:** 20 curated administrative queries; a single outcome change shifts performance by 5.0 percentage points (requires cautious statistical interpretation).
- **Corpus Scope:** 51 pages across 4 documents from a single university; 29 pages OCR-processed without complete token-level human audit.
- **Evaluation Methodology:** Correctness scores were assigned via automated GPT-based qualitative review; double-blind human cross-annotation remains pending.
- **Generator Scale:** Evaluated on a frozen 1.5B instruction-tuned model (`Qwen2.5-1.5B-Instruct`); larger models (7B/72B) may better interpret complex tables.
- **Inference Context Overhead:** Top-5 RAG increases prompt tokens from 236 to 2,749–3,205 tokens ($\approx 12\times$ latency and API cost increase).

CS420 S07 Report §7. Limitations documented in accordance with seminar evaluation standards.

Proposed Improvements: Verification & Maintenance

1. Post-Generation Verification Loop (Schneider, 2024 — GenXAI)

- **Atomic Claim Decomposition:** Deconstruct generated answers into verifiable tuples:

$$\langle \text{Claim},\ \text{Applicable Cohort},\ \text{Chunk ID},\ \text{Verbatim Quote} \rangle$$

- **Natural Language Inference (NLI) Gate:** Automatically audit entailment between claim and quote; trigger safe abstention ("Insufficient information") if ungrounded.

2. Tabular Structuring & Regulatory Maintenance

- **Structured Table Parsing:** Convert dense multi-column tuition matrices into explicit key-value schemas to eliminate column misalignment (as observed in Q13).
- **Versioned Corpus Registry:** Track circulars by decision number, effective date, and SHA-256 hash; execute automated diffs to retire obsolete regulatory chunks.

Schneider (2024), GenXAI, §3.3; CS420 S07 Report §6. Proposed architectural improvements.

Conclusions & Core Takeaways

- **Substantial Accuracy Lift:** RAG elevates core factual correctness from 0% in an unaugmented baseline to 45–50% on complex administrative regulations.
- **Complementary Retrieval Roles:** BM25 delivers highest top-1 precision (Recall@1 = 80.00%); BGE-M3 provides greatest aggregate coverage (Recall@5 = 89.17%).
- **The Verification Imperative:** High recall (Hit@5 = 95%) is insufficient. Generators omit conditions, confuse tables, and fail at citation IDs.
- **Core Takeaway:** Curing hallucinations requires treating retrieval, structured parsing, and post-generation claim verification as an integrated pipeline.

External documents ground LLMs, but only rigorous verification ensures reliability.

CS420 S07 Seminar Report, Faculty of Information Technology, HCMUS (2026).

Key References

- **Lewis, P. et al. (2020).** Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 9459–9474.
- **Schneider, J. (2024).** Explainable Generative AI (GenXAI): A survey, conceptualization, and research agenda. *Artificial Intelligence Review*, 57, 289.
- **Chen, J. et al. (2024).** M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. *arXiv:2402.03216*.
- **Qwen Team (2024).** Qwen2.5 Technical Report. *arXiv:2412.15115*.
- **VNU-HCM University of Science (2020–2026).** Decisions 3087/QĐ-KHTN (Academic), 3317/QĐ-KHTN (Tuition), 2464/QĐ-KHTN (Language), and Student Conduct 2020/2023.
- **Project Artifacts & Codebase:** `https://github.com/namkiet/CS420_SeminarRAG`

All references verified and archived in repository directory `docs/` and `report/`.

Thank you for listening!

Any questions?